CHAPTER 18

# Cell Death

The development and maintenance of multicellular organisms depend not only on cell growth and cell division but also on cell death. During animal development, for example, carefully orchestrated patterns of cell growth, division, and death help determine the size and shape of limbs and organs. The maintenance of tissue size in adult animals often requires that cells die at the same rate as they are produced—a process called cell turnover. Such "normal" cell death also occurs in plants during development and in the senescence of flowers and leaves, and it can also occur in unicellular organisms, including yeasts and bacteria. Most of these normal cell deaths are suicides, in which a sequence of molecular events destroys the cells from within, but the molecular mechanisms can differ widely. In this chapter, we focus on a molecularly distinct form of cell suicide called **apoptosis** (from the Greek word meaning "falling off," as leaves from a tree), even though it occurs only in animal cells. Apoptosis is thought to occur in all animals, and it is by far the most common way for our cells to die. It has also been the most intensely studied form of cell death.

A cell dying by apoptosis undergoes characteristic morphological and biochemical changes, in part, to package itself to be eaten and digested quickly by a nearby cell. **Figure 18–1** shows a normal rat cell (panel A) and rat cells of the same type undergoing two different forms of cell death (panels B–D). In apoptosis, the cell shrinks and condenses, the cytoskeleton collapses, the nuclear envelope disassembles, and the nuclear chromatin condenses and fragments (Figure 18–1B). The apoptotic cell surface often forms multiple large protrusions called blebs (**Movie 18.1**), and, if the cell is large, it breaks up into membrane-enclosed fragments called apoptotic bodies. Importantly, the surface of the cell and its fragments become chemically altered, so that a neighboring cell, often a macrophage in vertebrates, rapidly engulfs the cell and fragments before they can spill their contents (Figure 18–1C). In this way, the cell dies neatly, and the cell and its fragments are rapidly cleared away, without causing a damaging inflammatory response. Because apoptotic cells are eaten and digested so quickly, there are often few dead cells to be seen, even in a tissue where large numbers of cells have died by apoptosis.

Damaged or infected cells can also die by apoptosis, ensuring that they are eliminated before they can threaten the health of the animal. There are, however, many non-apoptotic ways for severely damaged or stressed animal cells to die. These are usually lumped together under the umbrella term *cell necrosis*. A common, largely passive form of cell necrosis occurs when cells lyse in response to an acute tissue insult, such as trauma or a blocked blood supply: the cells swell and burst (Figure 18–1D), spilling their contents over their neighbors and eliciting an inflammatory response. In other forms of cell necrosis, the cell takes an active part in the death process, and they are given special names such as necroptosis or pyroptosis, depending on the nature of the inducing stress and the molecular mechanisms involved.

In this chapter, we discuss the major functions of apoptosis in vertebrates, its molecular mechanism and regulation, and how excessive or insufficient apoptosis can contribute to human disease.

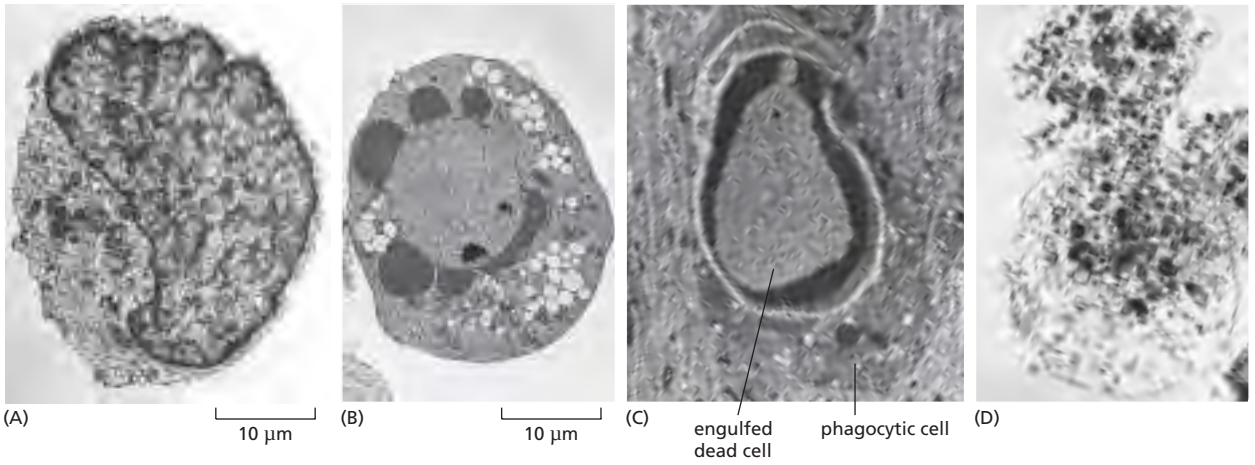


**Figure 18–1 Two distinct forms of cell death.** These electron micrographs show developing rat oligodendrocyte precursor cells in different states: (A) a normal cell in culture; (B) a cell in culture that has died by apoptosis because it was deprived of extracellular survival signals (discussed later); (C) a cell in a normal developing optic nerve that has died by apoptosis and has been engulfed by a neighboring phagocytic cell; (D) a cell in culture that died by necrosis. Note that the apoptotic cells in B and C have an intact plasma membrane, but the chromatin has become condensed and distorted and concentrated at the margin of the nucleus, whereas the necrotic cell in D seems to have exploded. The large vacuoles visible in the cytoplasm of the cell in B are a variable feature of apoptosis. (Courtesy of Julia Burne and Martin Raff.)

## Apoptosis Eliminates Unwanted Cells

The amount of apoptotic cell death that occurs in many developing and adult vertebrate tissues is astonishing: at least a million cells die this way each second in a healthy adult human (and are replaced by cell division). It seems remarkably wasteful for so many cells to die, especially as the vast majority are perfectly healthy at the time they kill themselves. What is the benefit of such massive cell death?

In some cases, especially in animal development, the function of cell death is clear. Apoptosis helps sculpt our hands and feet during embryonic development: these appendages start out as spade-like structures, and the individual digits separate only as the cells between them die, as illustrated for a mouse paw in **Figure 18–2**. In other cases, cells die by apoptosis when the structure they form is no longer needed. When a tadpole changes into a frog at metamorphosis, for example, the cells in the tail die by apoptosis, and the tail, which is not needed in the frog, disappears. Apoptosis also functions as a quality-control process in development, eliminating cells that are abnormal, misplaced, nonfunctional, or potentially dangerous to the animal. Striking examples occur in the vertebrate adaptive immune system, where apoptosis eliminates developing T and B lymphocytes that either fail to produce potentially useful antigen-specific receptors or produce self-reactive receptors that make the cells potentially dangerous (discussed in Chapter 24); it also eliminates most of the lymphocytes activated to proliferate by an infection, after they have helped destroy the responsible microbes.

In adult tissues that are neither growing nor shrinking, cell death and cell division must be tightly regulated to ensure they are in balance. If part of the liver is removed in an adult rat, for example, liver cell proliferation increases to make up the loss. Conversely, if a rat is treated with the drug phenobarbital—which stimulates liver cell growth and division (and thereby liver enlargement)—and then the phenobarbital treatment is stopped, apoptosis in the liver greatly increases until the liver cell number has returned to normal, usually within a week or so. Thus, liver cell number is kept constant through the regulation of both the cell death rate and the cell birthrate. The control mechanisms responsible for such remarkable regulation are largely unknown.

Animal cells can recognize damage in their various organelles and, if the damage is great enough, they can kill themselves by undergoing apoptosis. An important example is DNA damage, which can produce cancer-promoting

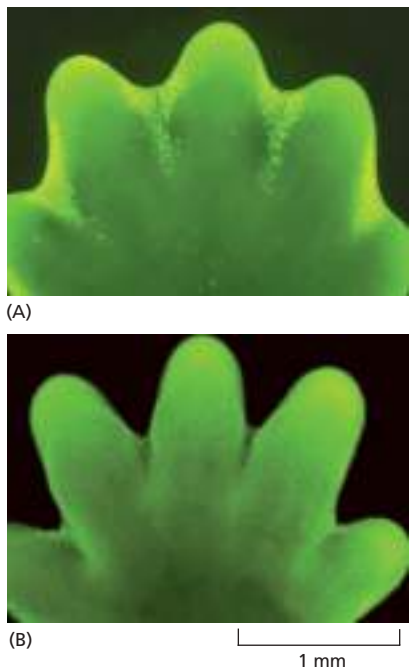


**Figure 18–2 Sculpting the digits in the developing mouse paw by apoptosis.** (A) The paw in this mouse fetus has been stained with the dye acridine orange, which enters apoptotic cells and thereby brightly labels them in the normal developing paw. The apoptotic cells appear as *bright green dots* concentrated between the developing digits. (B) The interdigital cell death has eliminated much of the tissue between the developing digits, as seen one day later, when there are very few apoptotic cells. (From W. Wood et al., *Development* 127:5245–5252, 2000. With permission from the Company of Biologists.)

mutations if not repaired. Cells have various ways of detecting DNA damage (see Figure 17–60) and can die by apoptosis if they cannot repair it.

## Apoptosis Depends on an Intracellular Proteolytic Cascade Mediated by Caspases

A family of specialized intracellular proteases triggers apoptosis by cleaving numerous, but specific, intracellular proteins at specific amino acid sequences, thereby bringing about the dramatic changes that occur during apoptosis. Because these proteases have a cysteine at their active site and cleave their target proteins at specific aspartic acids, they are called **caspases** (*c* for cysteine and *asp* for aspartic acid). Not all caspases are involved in apoptosis: indeed, the first human caspase identified, caspase-1, mainly helps stimulate inflammatory responses by cleaving the precursors of two pro-inflammatory, extracellular signal molecules (cytokines, discussed in Chapter 24). The caspases involved in apoptosis preexist in the cytosol of nearly all our cells as inactive precursors (often called procaspases), which are activated during apoptosis. There are two major classes of these apoptotic caspases: *initiator* caspases and *executioner* caspases.

**Initiator caspases**, as their name implies, begin the apoptotic program. In mammals, they are mainly *caspase-8* and *caspase-9*. They are made as inactive soluble monomers and are activated only when the monomers dimerize. The dimerization occurs when an apoptotic signal triggers the assembly of a specific *adaptor-protein* complex, which then recruits multiple copies of identical initiator caspase monomers to form larger *activation complexes*, within which the monomers dimerize and become activated. Each monomer in the activated caspase dimer then cleaves its partner at specific sites to form the mature, activated, initiator caspase dimer (**Figure 18–3**).

The major function of initiator caspases is to activate the **executioner caspases**, which orchestrate the apoptosis program. There are three executioner caspases in vertebrates—*caspase-3*, *caspase-6,* and *caspase-7*. Unlike initiator caspases, executioner caspases normally exist as inactive soluble dimers, which are activated by cleavage, almost always mediated by an initiator caspase (**Figure 18–4**). Each initiator caspase can activate many copies of one or more executioner caspases, resulting in an amplifying *caspase cascade*. Once activated, executioner caspases catalyze the widespread protein cleavage events that are responsible for killing the cell in a characteristic way and preparing it for rapid engulfment and digestion

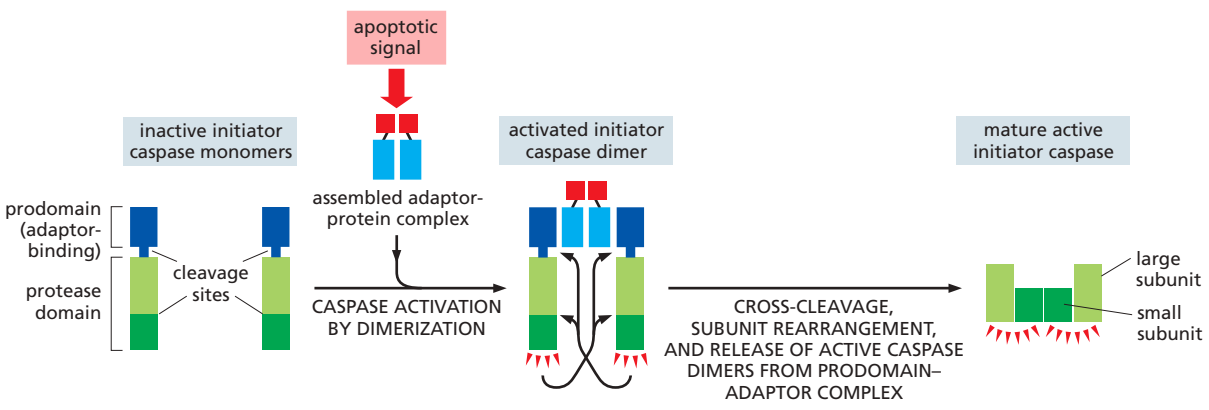


**Figure 18–3 Activation of an initiator caspase at the start of apoptosis.** An initiator caspase contains a protease domain in its large carboxyl-terminal region and a smaller, adaptor-protein-binding prodomain in its amino-terminal region. The caspase is made as an inactive monomer, which is activated by dimerization. The dimerization and activation only occurs when apoptotic signals trigger the assembly of specific adaptor-protein complexes (shown here as a hypothetical simplest form—a homodimer). Each type of adaptor complex then recruits multiple copies of one type of initiator caspase monomer, allowing them to dimerize and thereby become active. In the case shown, once activated, each monomer in the activated caspase dimer cross-cleaves its partner at specific sites; the cleavage in the protease domain enables the large and small caspase subunits to rearrange, and cleavage between the large protease domain and the prodomain releases the mature, active caspase dimer from the prodomain–adaptor complex.

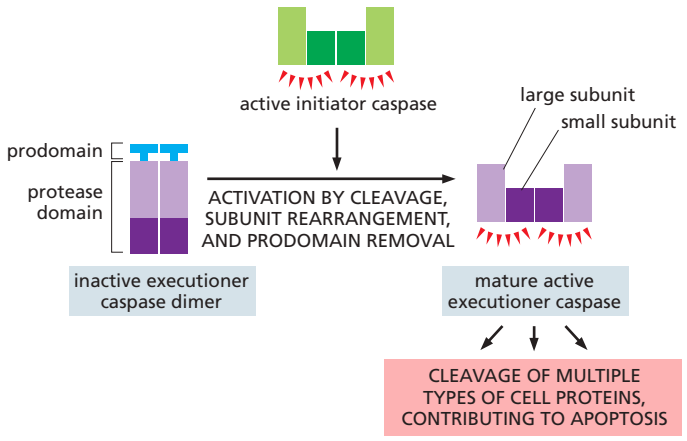


**Figure 18–4 Executioner caspase activation during apoptosis.** Executioner caspases have very short prodomains, which lack sites for interacting with other proteins. They are initially formed as inactive dimers, which are activated by cleavage at a site in each protease domain, almost always by an initiator caspase. The cleavages allow the large and small subunits to rearrange to form two active protease sites, each of which then cleaves off the prodomain of its partner monomer to produce the mature, active executioner caspase, as shown. The mature activated caspase then cleaves a variety of cell target proteins, leading to the controlled apoptotic death of the cell.

by neighboring cells. The caspase-initiated proteolytic cascade is not only self-amplifying and destructive but also irreversible, so that once a cell starts out along the path to apoptotic death, it cannot turn back.

Executioner caspases cleave hundreds of different cell proteins during apoptosis, but the roles of these target proteins in apoptosis are known in only a minority of cases, several of which we mention here. The cleavage of nuclear lamins by caspase-6, for example, causes the irreversible breakdown of the nuclear lamina (discussed in Chapter 12). The cleavage by caspase-3 of an inhibitor protein that normally holds a particular DNA-degrading endonuclease in an inactive form frees the endonuclease to cut up the DNA in the cell nucleus during apoptosis (**Figure 18–5**). The cleavage of certain proteins that regulate the actin cytoskeleton results in the actin polymerization in the cell cortex that is responsible for the surface blebbing in apoptosis, mentioned earlier (and see Movie 18.1). The cleavage of other actin regulators and some cell–cell adhesion proteins that attach cells to their neighbors helps an apoptotic cell round up and detach from its neighbors, making it easier for a neighboring cell to engulf it. As we discuss later, the cleavage of two phospholipid transfer proteins in the plasma membrane results in the exposure of phosphatidylserine on the surface of apoptotic cells, where it serves as an "eat me" signal to neighboring phagocytic cells. Importantly, preventing any one of these individual protein-cleavage steps does not stop apoptotic cell death, although it changes its characteristics as expected in each case.

How is an initiator caspase first activated in response to an apoptotic signal? Our cells use two main activation pathways: one is signaled from outside the cell and is called the *extrinsic pathway*, and the other is signaled from mitochondria

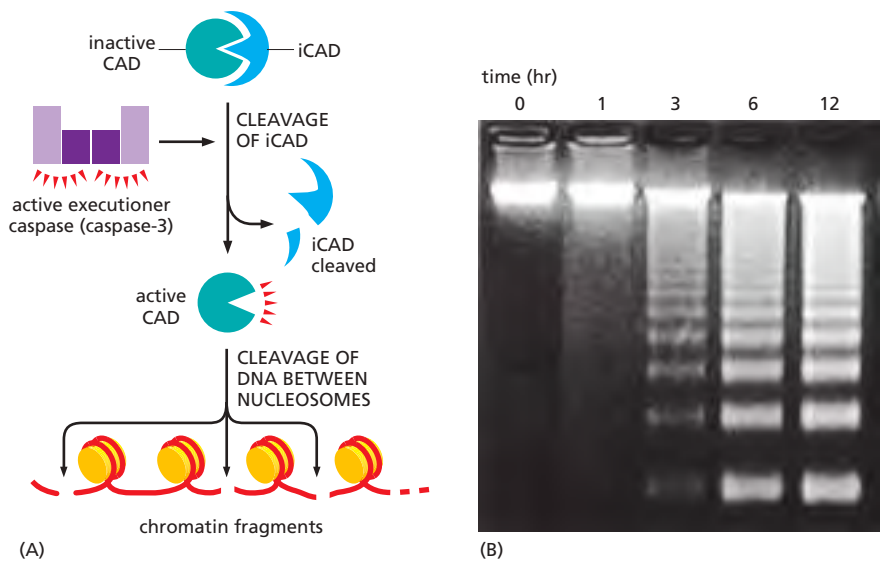


**Figure 18–5 DNA fragmentation during apoptosis.** (A) In healthy cells, a caspase-activated DNase (CAD) is held in an inactive state by an inhibitor protein, iCAD. Activation of an executioner caspase, caspase-3, during apoptosis leads to cleavage of iCAD, releasing the active DNase to cut the chromosomal DNA between nucleosomes. (B) Because the DNA cleavage occurs only at accessible sites in linker regions between nucleosomes, the DNA is cut into fragments of variable size, equivalent to the DNA associated with either one or multiple nucleosomes (as shown in A), producing a ladder pattern upon DNA gel electrophoresis. The pattern shown was obtained by inducing apoptosis in mouse thymus lymphocytes with dexamethasone, extracting DNA at the times indicated at the top of the gel, separating the fragments by size by electrophoresis in an agarose gel, and staining the DNA in the gel with ethidium bromide. (B, from D. McIlroy et al., *Genes Dev.* 14:549–558, 2000. With permission from Cold Spring Harbor Laboratory Press.)

inside the cell and is called the *intrinsic*, or *mitochondrial, pathway*. Each uses its own initiator caspase and adaptor proteins, as we now discuss.

## Activation of Cell-Surface Death Receptors Initiates the Extrinsic Pathway of Apoptosis

Extracellular signal proteins binding to cell-surface **death receptors** trigger the **extrinsic pathway** of apoptosis. Death receptors are transmembrane proteins containing an extracellular ligand-binding domain, a single transmembrane domain, and an intracellular *death domain* that is required for the receptors to activate the apoptotic program. The receptors are homotrimers and belong to the *tumor necrosis factor (TNF) receptor* family, which has eight members, including a receptor for TNF itself and the *Fas* death receptor. The ligands that activate the death receptors are also homotrimers, belonging to the *TNF family* of signal proteins.

A relatively well-understood example of how death receptors trigger the extrinsic pathway of apoptosis is the activation of the **Fas receptor** on the surface of a target cell by **Fas ligand** on the surface of a killer (cytotoxic) lymphocyte (discussed in Chapter 24). Fas signaling has a role in regulating the numbers of T and B lymphocytes, as indicated by the finding that inactivation of Fas signaling results in an abnormal increase in these cells. As shown in **Figure 18–6**, the binding of

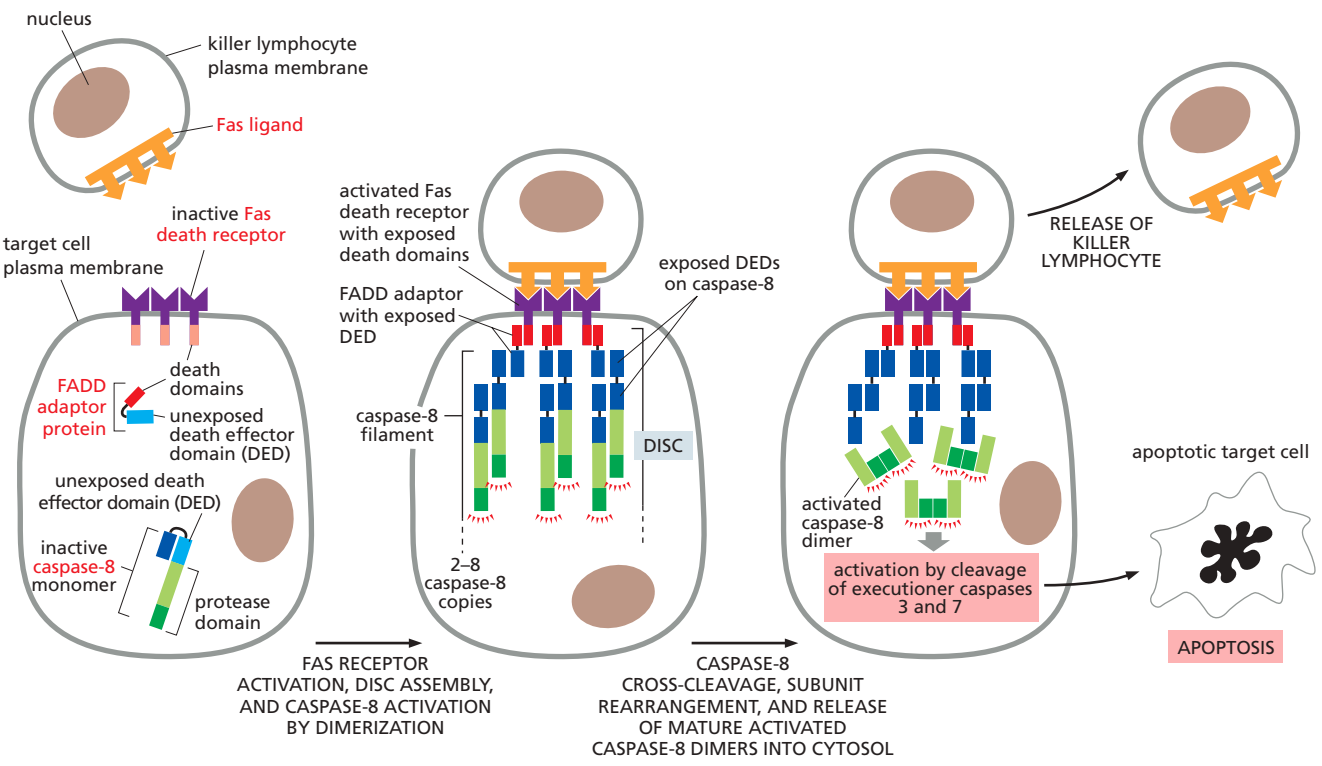


**Figure 18–6 The extrinsic pathway of apoptosis activated through Fas death receptors.** Trimeric Fas ligands on the surface of a killer lymphocyte bind to trimeric Fas death receptors on the surface of a target cell, inducing the target cell to kill itself by undergoing apoptosis by the extrinsic pathway. Although not shown, at least two trimeric Fas ligands have to bind to and cluster at least two trimeric Fas receptors to activate the pathway; for clarity, only a single copy of the ligand and receptor is shown. The ligand-induced receptor clustering (not shown) exposes a *death domain* on the receptor tails, as indicated here by the change in the color of the domain from *light red* to *dark red* upon exposure. Each exposed death domain binds to a similar exposed death domain on the cytosolic adaptor protein FADD (for Fas-associated death domain). The bound FADD protein then exposes a *death effector domain* (DED; *dark blue*), enabling FADD to recruit an inactive, monomeric initiator caspase (mainly caspase-8) by binding to an exposed DED *(dark blue)* on the prodomain of the caspase. Each caspase-8 monomer has two DEDs, and, when one binds to an exposed DED, the other becomes exposed (as indicated by the change in the DED color from *light blue* to *dark blue*) and can recruit another caspase-8 monomer; this results in a chain reaction in which the caspase-8 monomers oligomerize into a three-dimensional helical filament (not shown), with each FADD protein attached to up to eight caspase-8 molecules. The end result is the assembly of a large death-inducing signaling complex (DISC) composed of multiple copies of Fas, FADD, and caspase-8. Within the DISC, neighboring caspase-8 monomers interact to form activated dimers, which can now cross-cleave their partner monomers (not shown, but see Figure 18–3), cleaving off the prodomain and forming the mature activated dimers, which are released into the cytosol, where they can cleave and activate executioner caspases to induce apoptosis. In human cells, the caspase-10 initiator caspase can also be incorporated into the DISC, but, unlike caspase-8, it is not essential for Fas-induced apoptosis.

trimeric Fas ligands to the trimeric Fas receptors clusters the receptors, exposing death domains on the receptor tails, which then bind and cluster a small intracellular adaptor protein called *FADD*. The clustered FADD proteins, in turn, recruit multiple copies of an inactive, monomeric initiator caspase (mainly caspase-8), which oligomerize, completing the formation of a large **death-inducing signaling complex (DISC)** on the cytoplasmic face of the target-cell plasma membrane. The oligomerization of caspase-8 allows the dimerization and activation of the caspase, which then cleaves itself to form mature, active caspase-8 dimers that can cleave and activate downstream executioner caspases to induce apoptosis.

The sensitivity of different cell types to Fas-induced apoptosis varies. Many cells produce inhibitory proteins that act to restrain the extrinsic apoptotic pathway. Some cells, for example, produce a protein called *FLIP*, which resembles caspase-8 but lacks protease activity because it is missing the key cysteine required in the active site. FLIP dimerizes with caspase-8 in the DISC and prevents it from activating executioner caspases to initiate apoptosis. In this way, FLIP sets an inhibitory threshold that the extrinsic pathway, through activated caspase-8, must overcome to trigger apoptosis. As we discuss later, for the extrinsic pathway to kill some cell types, it has to overcome another caspase inhibitor, which it does by recruiting the intrinsic apoptotic pathway—the pathway we now discuss.

## The Intrinsic Pathway of Apoptosis Depends on Proteins Released from Mitochondria

Cells can also activate their apoptosis program from inside the cell, often in response to developmental signals or to injury such as DNA damage. In vertebrate cells, these apoptotic responses are mediated by the **intrinsic pathway** of apoptosis, which is also called the **mitochondrial pathway**, as it depends on the release into the cytosol of mitochondrial proteins that normally reside in the intermembrane space of these organelles (see Figure 12–47). The most important of these released proteins is **cytochrome *c***, which is a water-soluble component of the mitochondrial electron-transport chain and therefore has a central role in ATP production by oxidative phosphorylation in mitochondria (see Figure 14–18). When released into the cytosol (**Figure 18–7**), however, it takes on an entirely new function: it can induce apoptosis, independent of its electron-transport activity.

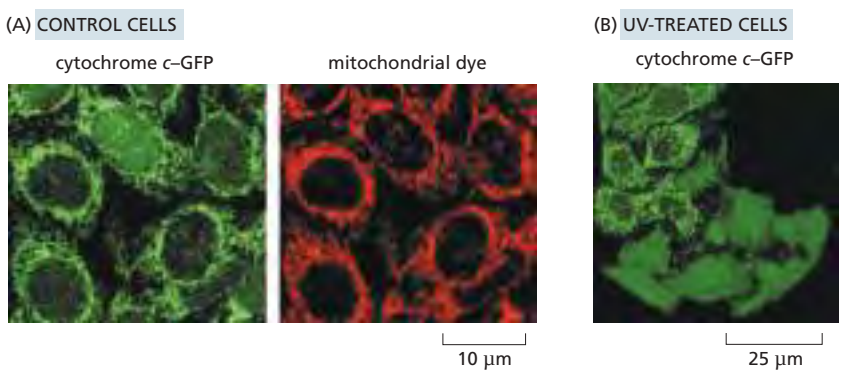


**Figure 18–7 Release of cytochrome *c* from mitochondria into the cytosol during the intrinsic pathway of apoptosis.** These fluorescence micrographs show human cancer cells in culture. The cells were transfected with a gene encoding a fusion protein consisting of cytochrome *c* linked to green fluorescent protein (cytochrome *c*–GFP), and they were also treated with a red dye that accumulates in mitochondria. (A) Unstimulated control cells: the overlapping distribution of the *green* and *red* confirms that the cytochrome *c*–GFP is located in mitochondria. (B) Cells were irradiated with ultraviolet (UV) light to induce the intrinsic pathway of apoptosis and were photographed 5 hours later. The seven cells in the bottom right of this micrograph have released their cytochrome *c* from mitochondria into the cytosol, whereas the other cells in the micrograph have not yet done so. (From J.C. Goldstein et al., *Nat. Cell Biol.* 2:156–162, published 2000 by Nature Publishing Group. Reprinted with permission of SNCSC.)

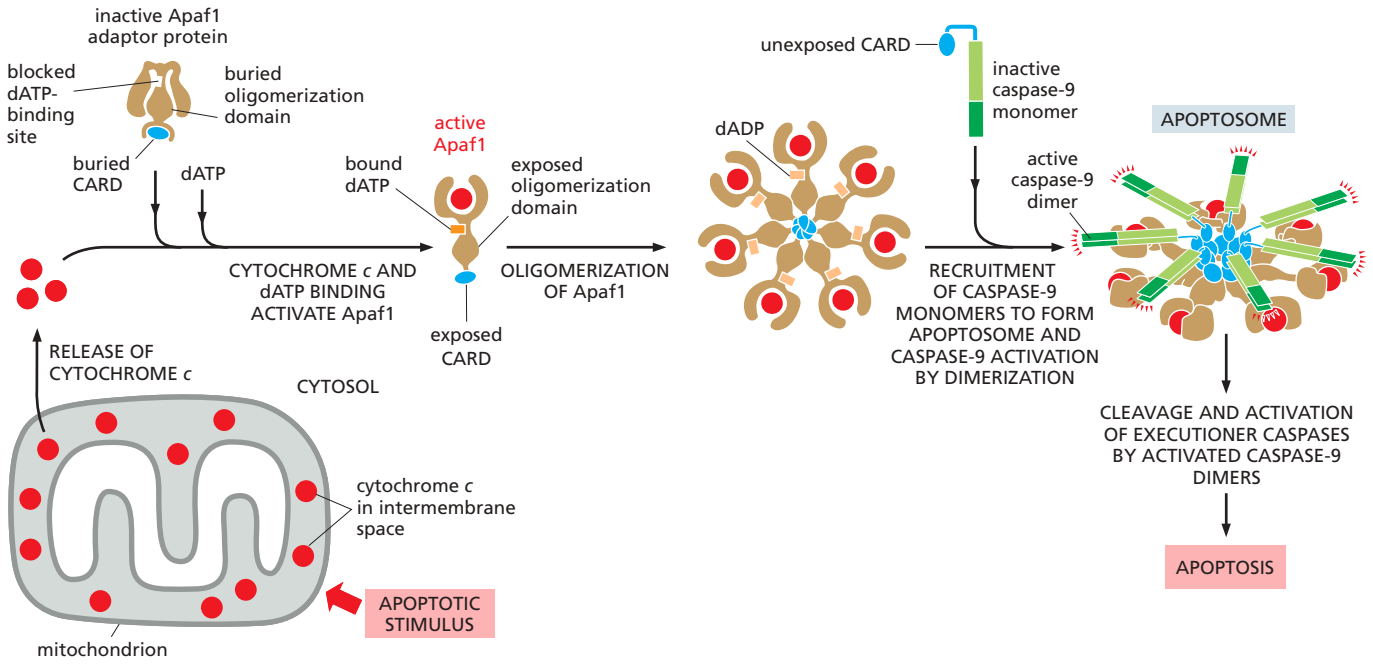


**Figure 18–8 The intrinsic pathway of apoptosis.** Intracellular apoptotic stimuli cause mitochondria to release cytochrome *c*. The binding of cytochrome *c* to the cytosolic adaptor protein Apaf1 induces a conformational change in Apaf1, which activates it, exposing a binding site for deoxy-ATP (dATP), an oligomerization domain, and a caspase recruitment domain (CARD). The exposed oligomerization domain mediates the assembly of Apaf1 into a wheel-like heptamer. Each CARD in the heptamer then recruits an inactive caspase-9 monomer through its own CARD, forming a large apoptosome, with the interacting CARDs clustered above the central hub of the apoptosome. The dATP is hydrolyzed to dADP during the assembly process. Within the apoptosome, the caspase-9 monomers are activated by dimerization. Activated caspase-9 dimers then cleave and activate downstream executioner caspases, leading to apoptosis. Note that the CARD domain is related in structure and function to the death effector domain (DED) of caspase-8 (see Figure 18–6).

(It remains an important mystery how, during evolution, mitochondria and cytochrome *c* came to acquire their surprising role in apoptosis.)

Once released, into the cytosol, cytochrome *c* binds to an adaptor protein called **Apaf1** (*apoptotic protease activating factor 1*), causing the adaptor to bind deoxy-ATP and oligomerize into a large, wheel-like heptamer. The heptamer then recruits inactive initiator caspase-9 monomers into the complex, forming an even larger structure called an **apoptosome** (**Figure 18–8**). Caspase-9 is activated by dimerization within the apoptosome, just as the other major initiator caspase, caspase-8, is activated by dimerization within the DISC (see Figure 18–6). Once activated, caspase-9 cleaves and activates the downstream executioner caspases that mediate apoptosis.

The intrinsic pathway is responsible for the great majority of apoptotic cell deaths in vertebrates. For upstream signals to activate the intrinsic pathway, they have to alter the outer mitochondrial membrane so that the soluble proteins in the intermembrane space such as cytochrome *c* can diffuse into the cytosol. This crucial outer membrane permeabilization step is controlled by interactions between members of the *Bcl2 family* of proteins, as we now discuss.

## Bcl2 Proteins Are the Critical Controllers of the Intrinsic Pathway of Apoptosis

The intrinsic pathway is tightly regulated to ensure that cells kill themselves by apoptosis only when it is appropriate. This regulation is largely the function of the **Bcl2 family** of proteins, which are named after the first family member described (B cell lymphoma-2), as explained later. In mammalian cells, complex interactions between these proteins control the permeabilization of the outer mitochondrial membrane and thereby govern the release into the cytosol of cytochrome *c* and

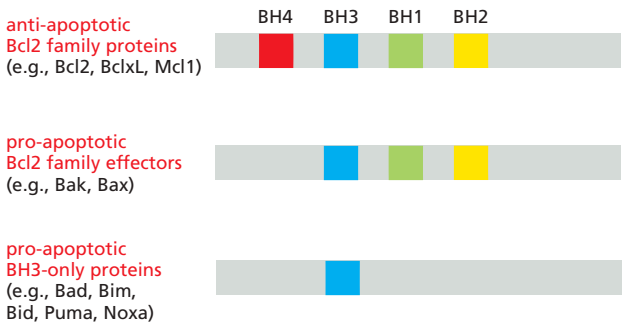


**Figure 18–9 Schematic drawing of the BH domains in the three classes of Bcl2 family proteins.** Note that the BH3 domain is the only one of the four BH domains shared by all Bcl2 family members; it mediates the direct interactions between pro-apoptotic and anti-apoptotic family members (see Figure 18–10C and D).

other soluble proteins in the intermembrane space, a process called **mitochondrial outer membrane permeabilization (MOMP)**. Like the caspase family, the Bcl2 family of proteins is found in all animals and has been remarkably conserved; a human Bcl2 protein, for example, can suppress apoptosis when expressed in the worm *Caenorhabditis elegans*.

There are three structural and functional classes of mammalian Bcl2 family proteins: (1) *Anti-apoptotic Bcl2 family proteins*, including Bcl2 itself, inhibit apoptosis by preventing MOMP. (2) *Pro-apoptotic Bcl2 family effectors* can directly induce MOMP by creating openings in the outer mitochondrial membrane. (3) A second class of pro-apoptotic Bcl2 family proteins, called *BH3-only proteins* (for reasons explained shortly), promotes apoptosis by regulating the other two classes. The balance between the activities of these three classes largely determines whether MOMP occurs or not and, therefore, whether a mammalian cell lives or dies by the intrinsic pathway of apoptosis.

As illustrated in **Figure 18–9**, the anti-apoptotic Bcl2 family proteins, including Bcl2 itself and *Bcl extra-large* (*BclxL*), share four distinctive *Bcl2 homology (BH) domains* (BH1–4). The two main pro-apoptotic Bcl2 family effectors, *Bak* and *Bax*, are structurally similar to Bcl2 but lack the BH4 domain. The members of the second class of pro-apoptotic Bcl2 family proteins share sequence homology with Bcl2 in only the BH3 domain and are therefore called *BH3-only proteins*; they are by far the largest class of Bcl2 proteins.

When an apoptotic stimulus triggers the intrinsic pathway, the **pro-apoptotic Bcl2 family effectors**, **Bak** and **Bax**, become activated and trigger MOMP by aggregating into oligomers of various sizes in the mitochondrial outer membrane, producing openings in the membrane by an uncertain mechanism, allowing cytochrome *c* and other intermembrane proteins to escape into the cytosol (**Figure 18–10A and B**). At least one of these pro-apoptotic effectors is required for the intrinsic pathway of apoptosis to operate in mammalian cells: mutant mouse cells that lack both proteins do not undergo MOMP or engage the intrinsic apoptotic pathway. Whereas Bak is bound to the mitochondrial outer membrane even in the absence of an apoptotic signal (see Figure 18–10A and B), Bax is mainly located in the cytosol until an apoptotic signal activates it, causing it to relocate to the outer membrane, where it oligomerizes. As we discuss below, the activation of Bak and Bax usually depends on activated BH3-only proteins.

The **anti-apoptotic Bcl2 family proteins** such as **Bcl2** and **BclxL** are also located on the cytosolic surface of the outer mitochondrial membrane, where they help prevent inappropriate MOMP. They do so by binding to the BH3 domain of active pro-apoptotic effectors Bak and Bax, thereby preventing their oligomerization (**Figure 18–10C**). There are at least five mammalian anti-apoptotic Bcl2 family proteins, and every mammalian cell requires at least one to avoid apoptosis and therefore to survive. Moreover, a number of these anti-apoptotic proteins must be inhibited for the intrinsic pathway to induce apoptosis, and BH3-only proteins mediate this inhibition.

The **BH3-only proteins**, the largest subclass of Bcl2 family proteins, promote MOMP and thereby apoptosis when they are either produced or activated in cells in response to an apoptotic stimulus. They do so in at least two ways. (1) Some BH3-only proteins, including Bad (see Figure 18–9), inhibit certain

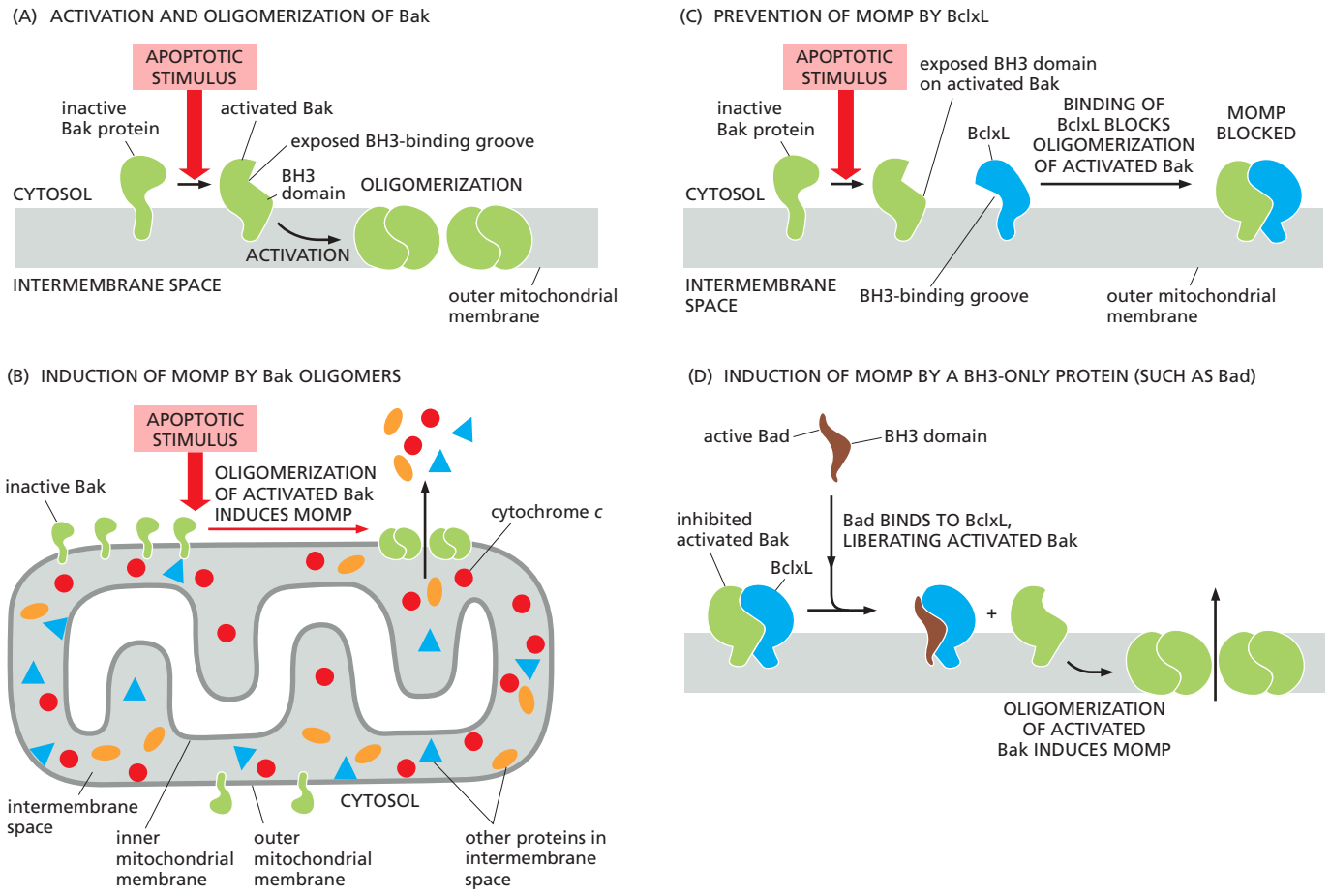


**Figure 18–10 How pro-apoptotic Bcl2 family effectors induce MOMP and how anti-apoptotic Bcl2 family proteins block it.** (A) Most of the pro-apoptotic effector Bak is already attached to the outer mitochondrial membrane before the protein is activated. When activated by an apoptotic stimulus, the protein undergoes a conformational change that both exposes a BH3 domain and creates a BH3-binding groove, allowing Bak–Bak oligomerization in the outer membrane. (B) The Bak oligomers induce MOMP by creating openings in the outer membrane that allow cytochrome *c* and other soluble proteins in the intermembrane space to diffuse into the cytosol; although the details of how Bak (and Bax) oligomers produce the openings are uncertain, it is thought that they form large ring structures that disrupt the integrity of the outer membrane. Once released into the cytosol, cytochrome *c* stimulates the assembly of apoptosomes (see Figure 18–8). (C) The anti-apoptotic Bcl2 family protein BclxL, like Bak, is normally bound to the outer mitochondrial membrane, where it can interact via its BH3-binding groove to the exposed BH3 domain on activated Bak, thereby blocking Bak–Bak oligomerization, MOMP, and apoptosis. (D) One way BH3-only proteins such as Bad are thought to indirectly induce MOMP and apoptosis is by inhibiting certain anti-apoptotic Bcl2 family proteins such as BclxL.

anti-apoptotic Bcl2 family proteins by binding via their BH3 domain to the BH3-binding groove on the anti-apoptotic protein. This binding blocks the anti-apoptotic activity of the Bcl2 family protein, thereby allowing Bax and/or Bak to oligomerize in the outer mitochondrial membrane to trigger MOMP (**Figure 18–10D**). (2) Some BH3-only proteins, including Bim and Bid (see Figure 18–9), can directly bind to and activate Bak and Bax, stimulating them to oligomerize and trigger MOMP.

In these ways, BH3-only proteins provide the crucial link between apoptotic stimuli and the intrinsic pathway of apoptosis, with different stimuli activating or inducing the production of different BH3-only proteins. When a cell suffers DNA damage that it cannot repair, for example, the tumor suppressor protein p53 (discussed in Chapters 17 and 20) accumulates in the nucleus and activates the transcription of genes that encode the BH3-only proteins *Puma* and *Noxa* (see Figure 18–9), which then trigger MOMP and apoptosis, thereby eliminating a potentially dangerous cell that could become cancerous. As we will see, some extracellular survival signals promote a cell's survival by preventing it from

undergoing apoptosis by inhibiting the synthesis or activity of certain BH3-only proteins (see Figure 18–13B).

As mentioned earlier, in some cells the extrinsic apoptotic pathway has to recruit the intrinsic pathway to help kill the target cell; the BH3-only protein *Bid* is the link between the two pathways. Bid is normally inactive in these cells, but, when death receptors activate the extrinsic pathway (see Figure 18–6), the initiator caspase, caspase-8, cleaves Bid, creating an active form of the protein that translocates to the outer mitochondrial membrane to activate Bax and/or Bak, causing them to oligomerize and trigger MOMP. MOMP is required for activated death receptors to kill these cells because it releases proteins from the mitochondrial intermembrane space that neutralize an inhibitory protein present in the cytosol of these cells that normally blocks apoptosis, as we explain next.

## An Inhibitor of Apoptosis (an IAP) and Two Anti-IAP Proteins Help Control Caspase Activation in the Cytosol of Some Mammalian Cells

Because activation of an apoptotic caspase cascade leads to certain cell death, cells employ multiple mechanisms to help ensure that these proteases are activated only when appropriate. The inhibition of caspase-8 by the FLIP protein in the extrinsic apoptotic pathway is one example discussed earlier (see p. 1095). Another line of defense against inappropriate caspase activation is provided by caspase inhibitor proteins called **inhibitors of apoptosis (IAPs)**. These proteins were first identified in certain insect viruses (baculoviruses), which encode IAP proteins to prevent a host cell that is infected by the virus from activating caspases and killing itself by apoptosis, which would curtail the virus's replication. Most animal cells also make IAP proteins, although most of these proteins do not regulate caspases and apoptosis.

Mammalian cells seem to have only one IAP that can directly inhibit caspase activity. It is called **XIAP** (because it is encoded on the X chromosome), and it resides in the cytosol of many of our cells. It binds to and inhibits caspase-9, an initiator caspase, and the executioner caspases, caspase-3 and caspase-7, thereby setting an inhibitory threshold that these caspases must overcome to trigger apoptosis. In these cells, XIAP also helps regulate the levels of the three caspase proteins: it has a ubiquitin-ligase domain that polyubiquitylates the caspases that XIAP binds to, marking them for destruction in proteasomes (see Figure 3–67).

When the intrinsic pathway of apoptosis is activated, among the proteins released from the mitochondrial intermembrane space by MOMP are two **anti-IAP proteins** called *Smac* and *Omi*. In cells with XIAP in their cytosol, these anti-IAP proteins bind to XIAP and prevent it from inhibiting caspases, thereby promoting caspase activation and apoptosis (**Figure 18–11**). This is why the extrinsic apoptotic pathway needs to recruit the intrinsic pathway (involving MOMP and caspases 9, 3, and 7) to induce apoptosis in some cells, as mentioned earlier; these are cells that express XIAP in their cytosol.

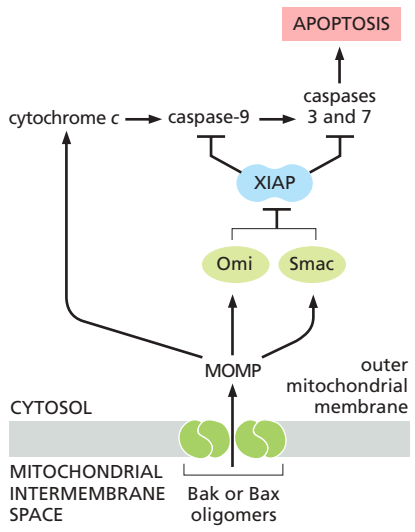


**Figure 18–11 How MOMP overcomes XIAP inhibition.** In some mammalian cells, the presence of XIAP in the cytosol inhibits initiator caspase-9 and executioner caspases 3 and 7; the XIAP binds to the active site of the three enzymes, blocking their activities. However, as well as releasing cytochrome *c*, MOMP releases two anti-IAP proteins, Smac and Omi, which inhibit the anti-caspase activity of XIAP, thereby allowing the activation of these caspases in the cytosol during the intrinsic pathway of apoptosis (see Figure 18–8).

## Extracellular Survival Factors Inhibit Apoptosis in Various Ways

As discussed in Chapters 15 and 21, extracellular signals regulate most activities of animal cells, including apoptosis. They are part of the normal "social" controls that ensure individual cells behave for the good of the organism as a whole—in this case, by surviving when the cells are needed and killing themselves when they are not. Some extracellular signal molecules stimulate apoptosis, whereas others inhibit it. We have discussed signal proteins such as Fas ligand that activate death receptors to trigger the extrinsic pathway of apoptosis. Other extracellular signal molecules that stimulate apoptosis are especially important during vertebrate development: a surge of thyroid hormone in the bloodstream, for example, signals cells in the tadpole tail to undergo apoptosis at metamorphosis. In mice, locally produced signal proteins stimulate cells between developing fingers and

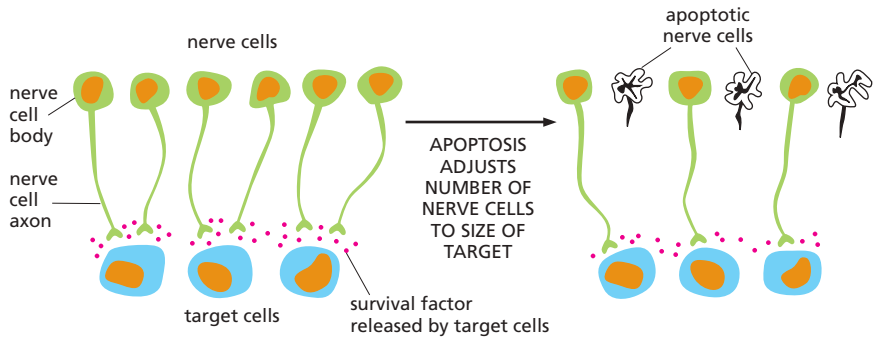


**Figure 18–12 How survival factors and apoptosis can help adjust the number of developing nerve cells to the size of the target tissue the nerve cells innervate.** In this example, more nerve cells are produced than can be supported by the limited amount of survival factors produced by the cells in a target tissue. As a result, some nerve cells receive an insufficient amount of survival factor to avoid apoptosis. This strategy of overproduction followed by culling during development helps ensure that all the appropriate target cells are contacted by appropriate nerve cells and that surplus nerve cells are automatically eliminated.

toes to kill themselves, thereby sculpting these digits (see Figure 18–2). Here, however, we focus on extracellular signal molecules that promote cell survival by inhibiting apoptosis and are collectively called **survival factors**.

Most animal cells require such signals from other cells to avoid undergoing apoptosis, usually by the intrinsic pathway. This surprising arrangement apparently helps ensure that cells survive only when and where they are needed. Some nerve cells, for example, are produced in excess in the developing nervous system and then compete for limited amounts of survival factors that are secreted by the target cells they normally connect to. Nerve cells that receive enough survival signals live, while the others die by apoptosis. In this way, the number of surviving neurons is automatically adjusted so that it is appropriate for the number of target cells they connect with (**Figure 18–12**). A similar competition for limited amounts of survival factors produced locally or systemically might help control cell numbers in other tissues, both during development and in adulthood.

Survival factors usually bind to cell-surface receptors, which activate intracellular signaling pathways (discussed in Chapter 15) that suppress the apoptotic program, usually by regulating the expression or activity of members of the Bcl2 family of proteins. Some survival factors, for example, stimulate the synthesis of anti-apoptotic Bcl2 family proteins such as Bcl2 itself or BclxL (**Figure 18–13A**). Some others act by inhibiting the function of pro-apoptotic BH3-only proteins such as Bad (**Figure 18–13B**). Some developing neurons, like those illustrated in Figure 18–12, use a counterintuitive alternative strategy, in which survival-factor receptors stimulate apoptosis when they are unoccupied and stop doing so when survival factors bind. The end result in all these cases is the same: cell survival depends on survival-factor binding to cell-surface receptors.

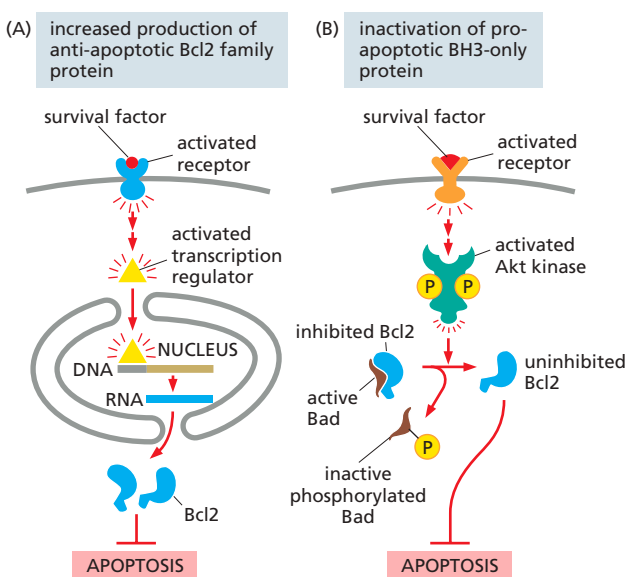


**Figure 18–13 Two of the various ways that extracellular survival factors can inhibit apoptosis.** (A) Some survival factors suppress apoptosis by stimulating the transcription of genes that encode anti-apoptotic Bcl2 family proteins such as Bcl2 (as shown here) or BclxL. (B) Many others activate the serine/threonine protein kinase Akt, which, among many other targets, phosphorylates and inactivates the pro-apoptotic BH3-only protein Bad (see Figure 18–9). When not phosphorylated, Bad promotes apoptosis by binding to and inhibiting an anti-apoptotic Bcl2 family protein such as Bcl2 itself. Once phosphorylated, Bad dissociates, freeing Bcl2 to suppress apoptosis (see Figure 18–10C and Figure 15–54). Akt can also suppress apoptosis by phosphorylating and inactivating transcription regulatory proteins that stimulate the transcription of genes encoding proteins that promote apoptosis, such as the BH3-only protein Bim (not shown). There are also many other ways that survival factors can inhibit apoptosis that are not illustrated here.

## Healthy Neighbors Phagocytose and Digest Apoptotic Cells

Apoptotic cell death is a tidy process. The apoptotic cell and its fragments do not break open and release their contents. Instead, they usually remain intact until they are rapidly phagocytosed and digested by neighboring cells (usually macrophages in humans), leaving no trace. In this way, apoptosis avoids triggering a destructive inflammatory response. The engulfment process depends on chemical changes on the surface of the apoptotic cell, which displays various "eat me" signals that are recognized by the phagocytic cells.

The most important of these signals is the negatively charged phospholipid **phosphatidylserine**. In healthy cells, phosphatidylserine is normally located exclusively in the inner leaflet of the lipid bilayer of the plasma membrane (see Figure 10–15); it is kept there by a specific *phospholipid flippase* that uses ATP hydrolysis to flip phosphatidylserine (PS) and phosphatidylethanolamine (PE) from the outer to the inner leaflet. In cells undergoing apoptosis, however, PS accumulates on the cell surface by two mechanisms. First, executioner caspases cleave the phospholipid flippase, thereby inactivating it, preventing PS and PE from being transferred from the outer to the inner leaflet. Second, executioner caspases cleave and thereby activate a *phospholipid scramblase* that transfers plasma membrane phospholipids nonspecifically between the inner and outer lipid leaflets, thereby scrambling them between the two leaflets. These two mechanisms responsible for the exposure of PS on the surface of apoptotic cells are illustrated in **Figure 18–14** and **Movie 18.2**. The appearance of PS on the cell surface also occurs in some forms of necrotic cell death, even in the absence of caspase activation; disruption of the plasma membrane is sufficient to do it.

The PS on apoptotic and necrotic cells is recognized by a variety of soluble "bridging" proteins that interact with both the exposed PS and specific receptors on the surface of a neighboring phagocytic cell, triggering the cytoskeletal and other changes that initiate the engulfment process.

Macrophages are professional phagocytes and will phagocytose dead cells, microbes, cell debris, latex beads, and almost any other particles, but they do not phagocytose healthy cells within the organism's own body—even those healthy cells that transiently expose phosphatidylserine on their surface when activated, such as platelets and some T lymphocytes. One reason is that almost all of our cells display signal proteins on their surface that bind to inhibitory receptors on the surface of macrophages, stimulating the receptors to block phagocytosis. Thus, in addition to expressing cell-surface "eat me" signals such as phosphatidylserine that stimulate phagocytosis, apoptotic cells must also remove or inactivate the "don't eat me" signals that block phagocytosis.

## Either Excessive or Insufficient Apoptosis Can Contribute to Disease

There are many human disorders in which too many cells undergo apoptosis and thereby contribute to pathological tissue loss. Among the most common and dramatic examples are heart attacks and strokes caused by an acute interruption of the blood supply to the heart or brain, respectively. In these conditions, many cells initially die by necrosis. If the blocked blood vessel is unblocked, either through an arterial catheter or with clot-disrupting drugs, some of the surviving cells in the oxygen-deprived tissue die by apoptosis, contributing to the tissue loss. It is hoped that, in the future, drugs that can block the apoptotic cell deaths, as well as drugs that can block some forms of necrotic cell death, will be able to decrease the tissue loss and its debilitating consequences.

There are other disorders in which too few cells die by apoptosis. For example, as mentioned earlier, mutations in mice and humans that inactivate the genes that encode either the Fas death receptor or its ligand prevent the normal deaths of some types of lymphocytes, resulting in the abnormal accumulation of these cells in the spleen and lymph glands. In many cases, this leads to autoimmune disorders, because some self-reactive lymphocytes fail to be eliminated and react

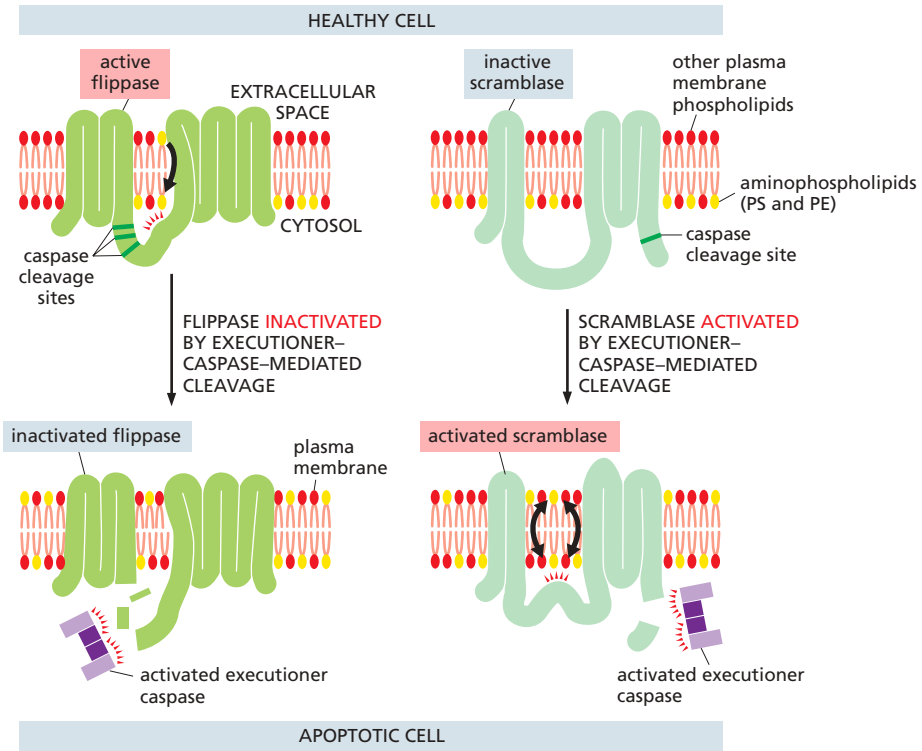


**Figure 18–14 The two caspase-dependent mechanisms responsible for the accumulation of phosphatidylserine on the surface of apoptotic cells.** In healthy cells *(top half of figure)*, an ATP-dependent aminophospholipid flippase (ATP11C) actively flips phosphatidylserine (PS) and phosphatidylethanolamine (PE) from the outer to the inner leaflet of the plasma membrane lipid bilayer, keeping these lipids mainly confined to the inner leaflet. During apoptosis *(bottom half of figure)*, activated executioner caspases (caspase-3, caspase-7, or both) cleave and thereby *inactivate* the ATP11C flippase, preventing its lipid translocation activity; at the same time, the activated executioner caspases cleave and thereby *activate* a phospholipid scramblase (Xkr8) in the plasma membrane that nonspecifically flips phospholipids between the two lipid leaflets of the membrane, scrambling them between the leaflets. (Although not shown, the ATP11C flippase is tightly associated with a smaller transmembrane protein, CDC50, which is required to chaperone the flippase to the plasma membrane and possibly to assist in the lipid translocation process.)

With the flippase permanently inactivated and the scramblase permanently activated, PS and PE become rapidly and irreversibly exposed on the apoptotic cell surface, where the PS serves as an "eat me" signal to neighboring phagocytic cells. Both of these caspase-dependent mechanisms are required to create this rapid and effective "eat me" signal: if only the flippase were inactivated, it could take a very long time for PS to accumulate in effective amounts on the apoptotic cell surface, because phospholipids rarely flip spontaneously between the two leaflets of a lipid bilayer without an enzyme (such as the scramblase) catalyzing it (see Figure 10–10B). Similarly, if only the scramblase were activated, the active flippase would rapidly return any scrambled PS from the external leaflet to the internal one, thereby removing the PS from the cell surface. Figure 12–39 illustrates the actions of phospholipid scramblases in lipid bilayer synthesis in the ER membrane and of phospholipid scramblases in generating lipid bilayer asymmetry in some other intraceullar membranes. (Modified from S. Nagata et al., *Cell Death Differ.* 23:952–961, 2016.)

against the individual's own tissues. The increase in lymphocytes can also lead to lymphocyte cancers called lymphomas.

Indeed, decreased apoptosis makes an important contribution to many types of cancer, because the normal inhibitory controls on apoptosis are often defective in cancer cells. The *Bcl2* gene, for example, was first identified in a common form of human lymphoma, in which a chromosome translocation causes excessive production of the anti-apoptotic Bcl2 protein (as mentioned earlier, Bcl2 gets its name from this *B cell lymphoma*). The increased amount of Bcl2 protein in the lymphocytes that carry the translocation promotes the development of cancer by inhibiting apoptosis, thereby abnormally prolonging the cells' survival and increasing their number; the increase in Bcl2 also decreases the cells' sensitivity to anticancer drugs, which often work by causing cancer cells to undergo apoptosis (discussed in Chapter 20).

Similarly, the gene encoding the tumor suppressor protein p53 is mutated in about 50% of human cancers so that it no longer promotes apoptosis or cell-cycle arrest in response to DNA damage (as discussed in Chapters 17 and 20). The lack of p53 function, therefore, enables the cancer cells to survive and proliferate even when their DNA is damaged; in this way, the cells progressively accumulate more mutations, some of which make the cancer more malignant. As many anticancer drugs induce apoptosis (and cell-cycle arrest) by p53-dependent mechanisms, the loss of p53 function also makes cancer cells less sensitive to these drugs.

If decreased apoptosis contributes to a cancer, then we should be able to treat the cancer with drugs that promote apoptosis. This approach has recently led to the development of small chemicals that block the function of anti-apoptotic Bcl2 family proteins such as Bcl2 and BclxL, by binding with high affinity to the BH3-binding groove of these proteins, in much the same way that pro-apoptotic BH3-only proteins do (see Figure 18–10D). These BH3-mimetic drugs (**Figure 18–15**) activate the intrinsic pathway of apoptosis, increasing the amount of tumor cell death in certain cancers, especially those that depend on particular anti-apoptotic Bcl2 family proteins for their survival.

Most human cancers are carcinomas, which arise in epithelial tissues such as those in the lung, intestinal tract, breast, and prostate (discussed in Chapter 20). Such epithelial cancer cells display many abnormalities in their behavior, including a decreased ability to adhere to the extracellular matrix and to one another at specialized cell–cell junctions. In the next chapter, we discuss these vitally important structures, which are responsible for holding our cells in tissues and organs in their proper place.

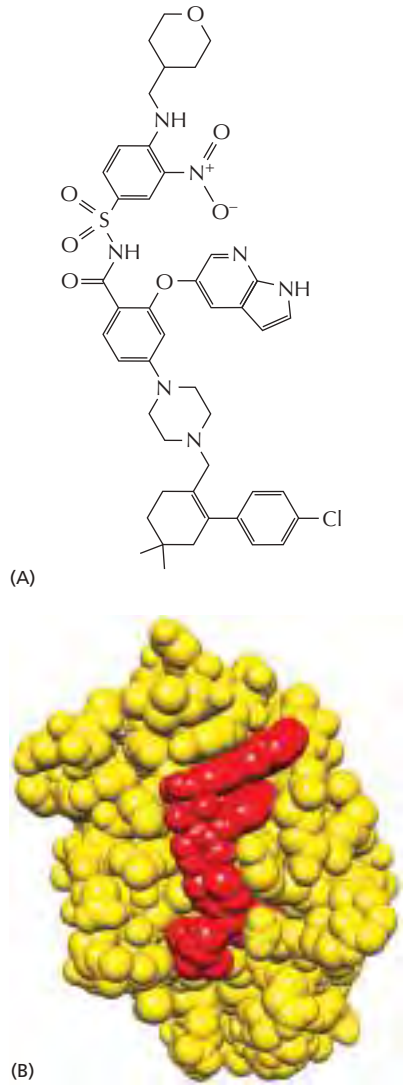


**Figure 18–15 A BH3-mimetic drug that specifically inhibits the Bcl2 anti-apoptotic Bcl2 family protein.** As shown in Figure 18–10D, one way BH3-only proteins can promote apoptosis is by binding to the long BH3-binding groove in anti-apoptotic Bcl2 family proteins such as Bcl2 itself, thereby preventing the protein from blocking apoptosis. (A) The chemical structure of *venetoclax*, which was designed and synthesized to bind tightly and specifically in the BH3-binding groove of Bcl2. (B) Crystal structure of venetoclax *(red)* bound to the human Bcl2 protein *(yellow)*. By inhibiting the activity of Bcl2, the drug promotes apoptosis in any cell that depends on this protein for survival, as is the case for cells of human chronic lymphocytic leukemia, for which this drug is in clinical use. (PDB code: 6OOK.)

## Summary

*Animal cells can activate various intracellular death programs and kill themselves when they are seriously damaged or stressed, no longer needed, or are a threat to the organism. In most cases, these deaths occur by apoptosis, in which the cells shrink, the nucleus and cells condense and often fragment, and neighboring phagocytic cells rapidly engulf the cells or fragments before there is any leakage of cytoplasmic contents. Apoptosis is mediated by proteolytic enzymes called caspases, which cleave specific intracellular proteins to kill the cell quickly and neatly.*

*Apoptotic caspases are present as inactive precursors in almost all nucleated animal cells. The activation of initiator caspases occurs when an apoptotic stimulus activates adaptor proteins, which bring inactive initiator caspase monomers into proximity within large activation complexes, in which the monomers are activated by dimerization. The activated initiator caspase dimers cleave themselves and then activate downstream executioner caspase dimers by cleaving them; the activated executioner caspases then cleave hundreds of target proteins in the cell. The amplifying, irreversible caspase cascade is responsible for all the events of apoptosis, including those that collectively kill the cell and prepare it for being phagocytosed and rapidly digested by a neighboring cell.*

*Cells use two distinct pathways to activate initiator caspases to trigger apoptosis: the extrinsic pathway is activated by extracellular ligands binding to cell-surface death receptors; the intrinsic pathway is activated from within the cell by developmental signals or stress signals. Each pathway uses its own initiator caspases, adaptor proteins, and activation complexes. In the extrinsic pathway, the death receptors recruit caspase-8 via adaptor proteins to form the activation complex called the DISC. In the intrinsic pathway, intracellular signals induce mitochondrial outer membrane permeabilization (MOMP), which releases soluble proteins from the mitochondrial intermembrane space into the cytosol; released cytochrome* c *activates the adaptor protein Apaf1, which recruits caspase-9 monomers to form a large activation complex called an apoptosome. Anti-apoptotic and pro-apoptotic intracellular Bcl2 family proteins interact with one another to tightly control MOMP to ensure that the intrinsic pathway of apoptosis is normally only activated when the death of the cell benefits the animal.*

# PROBLEMS

**Which statements are true? Explain why or why not.**

**18–1** In normal adult tissues, cell death usually balances cell division.

**18–2** Mammalian cells that do not have cytochrome *c* should be resistant to apoptosis induced by DNA damage.

**Discuss the following problems.**

**18–3** Fas ligand is a homotrimeric plasma membrane protein on killer lymphocytes that binds to a homotrimeric death receptor, Fas, on the surface of target cells, including some lymphocytes (**Figure Q18–1**). The clustering of trimeric Fas by the binding of clusters of Fas ligand alters the conformation of Fas so that it binds an adaptor protein, which then recruits and activates caspase-8, triggering a caspase cascade that leads to apoptotic cell death.

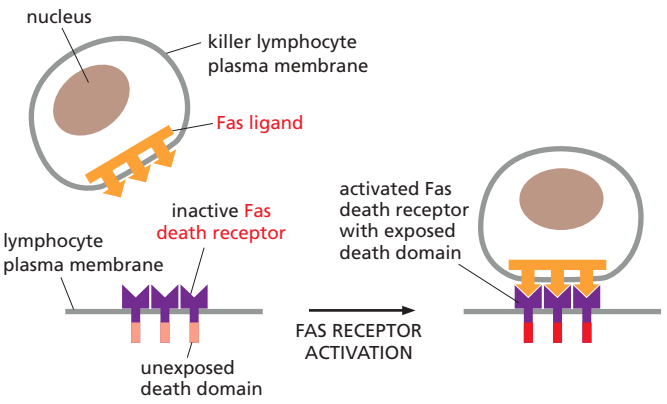


**Figure Q18–1** The binding of trimeric Fas ligand to Fas (Problem 18–3). Although not shown, at least two trimeric Fas ligands have to bind to and cluster at least two trimeric Fas receptors to activate the pathway; for clarity, only a single copy of the ligand and receptor is shown.

In humans, the autoimmune lymphoproliferative syndrome (ALPS) is associated with dominant mutations in Fas that include point mutations and C-terminal truncations. In individuals that are heterozygous for such mutations, some lymphocytes do not die at their normal rate and accumulate in abnormally large numbers, causing a variety of clinical problems. In contrast to these patients, individuals heterozygous for mutations that eliminate Fas expression entirely do not have these clinical problems.

Assuming that the normal and dominant forms of Fas are expressed to the same level and assemble randomly into trimers, what fraction of Fas–Fas ligand complexes on a lymphocyte from a heterozygous ALPS patient would be expected to be composed entirely of normal Fas subunits? Does your calculation suggest an explanation for why individuals heterozygous for expressed Fas mutants have clinical problems, whereas heterozygous individuals with unexpressed Fas mutants have no clinical problems?

**18–4** In contrast to their similar brain abnormalities, newborn mice deficient in Apaf1 or caspase-9 have distinctive abnormalities in their paws. Apaf1-deficient mice fail to eliminate the webs between their developing digits, whereas caspase-9–deficient mice have normally formed digits (**Figure Q18–2**). If Apaf1 and caspase-9 function in the same apoptotic pathway, how is it possible for these deficient mice to differ in web-cell apoptosis?

**Figure Q18–2** Appearance of paws in *Apaf1*$^{-/-}$ and *Casp9*$^{-/-}$ newborn mice relative to normal newborn mice (Problem 18–4). (From H. Yoshida et al., *Cell* 94:739–750, 1998. With permission from Elsevier.)

**18–5** When human cancer cells are exposed to ultraviolet (UV) light at 90 mJ/cm$^2$, most of the cells undergo apoptosis within 24 hours. Release of cytochrome *c* from mitochondria can be detected as early as 6 hours after exposure of a population of such cells to UV light, and it continues to increase for more than 10 hours thereafter. Does this mean that individual cells slowly release their cytochrome *c* over this time period? Or, alternatively, do individual cells release their cytochrome *c* rapidly but with different cells being triggered at variable times?

To answer this fundamental question, you have fused the gene for green fluorescent protein (GFP) to the gene for cytochrome *c*, so that you can observe the behavior of individual cells by confocal fluorescence microscopy. In cells that are expressing the cytochrome *c*–GFP fusion protein, fluorescence shows the punctate pattern typical of mitochondrial proteins. You then irradiate these cells with UV light and observe individual cells for changes in the punctate pattern. Two such cells (outlined in white) are shown in **Figure Q18–3A and B**. Release of

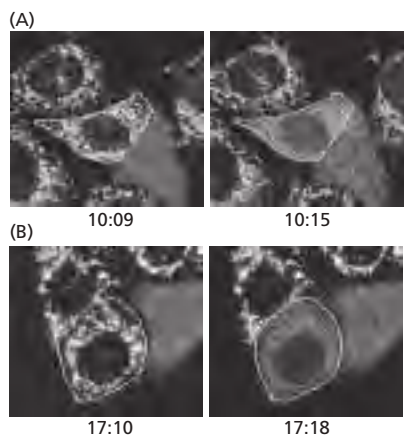


**Figure Q18–3** Analysis of cytochrome *c*–GFP release from mitochondria of individual cells by time-lapse video fluorescence microscopy (Problem 18–5). (A) Cells observed for 6 minutes, 10 hours after UV irradiation. (B) Cells observed for 8 minutes, 17 hours after UV irradiation. One cell in A and one in B, each *outlined in white*, have released their cytochrome *c*–GFP during the time frame of the observation, which is shown as hours:minutes below each panel. (From J.C. Goldstein et al., *Nat. Cell Biol.* 2:156–162, published 2000 by Nature Publishing Group. Reproduced with permission from SNCSC.)

cytochrome *c*–GFP is detected as a change from a punctate to a diffuse pattern of fluorescence characteristic of a cytosolic protein. Times after UV exposure are indicated as hours:minutes below the individual panels.

Which model for cytochrome *c* release do these observations support? Explain your reasoning.

**18–6** Imagine that you could microinject cytochrome *c* into the cytosol of wild-type mammalian cells, or into the cytosol of cells that were defective for the process of mitochondrial outer membrane permeabilization (MOMP). Would you expect the injected wild-type cells or the injected MOMP-defective cells to undergo apoptosis? Explain your reasoning.

**18–7** Which one of the following statements about Bcl2 family members is correct?

**A.** Bak is pro-apoptotic and Bax is anti-apoptotic.

**B.** Bax is pro-apoptotic and BclxL is anti-apoptotic.

**C.** Bcl2 is pro-apoptotic and Bak is anti-apoptotic.

**D.** BclxL is pro-apoptotic and Bcl2 is anti-apoptotic.

**18–8** One important role of Fas and Fas ligand is to mediate the elimination of tumor cells by killer lymphocytes. In a study of 35 primary lung and colon tumors, half the tumors were found to have amplified and overexpressed a gene for a secreted protein that binds to Fas ligand. How do you suppose that overexpression of this protein might contribute to the survival of these tumor cells? Explain your reasoning.

## REFERENCES


Adams JM & Cory S (2017) The BCL-2 arbiters of apoptosis and their growing role as cancer targets. *Cell Death Differ.* 25, 27–36.

Birkinshaw RW & Czabotar PE (2017) The BCL-2 family of proteins and mitochondrial outer membrane permeabilization. *Semin. Cell Dev. Biol*. 72, 152–162.

Czabotar PE, Lessene G, Strasser A & Adams JM (2014) Control of apoptosis by the BCL-2 protein family: implications for physiology and therapy. *Nat. Rev. Mol. Cell Biol.* 5, 49–63.

Danial NN & Korsmeyer SJ (2004) Cell death: critical control points. *Cell* 116, 205–219.

Datta SR, Ranger AM, Lin MZ . . . Greenberg ME (2002) Survival factor-mediated BAD phosphorylation raises the mitochondrial threshold for apoptosis. *Dev. Cell* 3, 631–643.

Davies AM (1996) The neurotrophic hypothesis: where does it stand? *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 351, 389–394.

Dorstyn L, Akey WA & Kumar S (2018) New insights into apoptosome structure and function. *Cell Death Differ.* 25, 1194–1208.

Ellis RE, Yuan JY & Horvitz HR (1991) Mechanisms and functions of cell death. *Annu. Rev. Cell Biol.* 7, 663–698.

Fu T-M, Li Y, Lu A . . . Wu H (2016) Cryo-EM structure of caspase-8 tandem DED filament reveals assembly and regulation mechanisms of the death-inducing signaling complex. *Mol. Cell* 64, 236–250.

Galluzzi L, Vitale I, Aaronson SA . . . Kroemer G (2018) Molecular mechanisms of cell death: recommendations of the Nomenclature Committee on Cell Death. *Cell Death Differ.* 25, 486–541.

Green DR (2018) Cell Death: Apoptosis and Other Means to an End, 2nd ed. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.

Ha HJ, Chun HL & Park HH (2020) Assembly of platforms for signal transduction in the new era: dimerization, helical filament assembly, and beyond. *Exp. Mol. Med*. 52, 356–366.

Jiang X & Wang X (2004) Cytochrome C-mediated apoptosis. *Annu. Rev. Biochem*. 73, 87–108.

Jost PJ, Grabow S, Gray D . . . Kaufmann T (2009) XIAP discriminates between type I and type II FAS-induced apoptosis. *Nature* 460, 1935–1039.

Julien O & Wells JA (2017) Caspases and their substrates. *Cell Death Differ.* 24, 1380–1389.

Jullien M, Gomez-Bougie, Chiron D & Touzeau C (2020) Restoring apoptosis with BH3 mimetics in mature B-cell malignancies. *Cells* 9, 717.

Kale J, Osterlund EJ & Andrews DW (2018) BCL-2 family proteins: changing partners in the dance towards death. *Cell Death Differ.* 25, 65–80.

Kalkavan H & Green DR (2018) MOMP, cell suicide as a BCL-2 family business. *Cell Death Differ.* 25, 46–55.

Kerr JF, Wylie AH & Currie AR (1972) Apoptosis: a basic biological phenomenon with wide-ranging implications in tissue kinetics. *Br. J. Cancer* 26, 239–257.

Mandal R, Barron JC, Kostova I . . . Strebhardt K (2020) Caspase-8: the double-edged sword. *Biochim. Biophys. Acta Rev. Cancer* 1873, 1–18.

Nagata S (2005) DNA degradation in development and programmed cell death. *Annu. Rev. Immunol*. 23, 853–875.

Nagata S, Suzuki J & Fujii T (2016) Exposure of phosphatidylserine on the cell surface. *Cell Death Differ*. 23, 952–961.

Silke J & Vince J (2017) IAPs and cell death. *Curr. Top. Microbiol. Immunol.* 403, 96–117.

Voss AK & Strasser A (2020) The essentials of developmental apoptosis. *F1000Res.* 9, F1000 Faculty Rev-148.